

TechMicra ERP — Final Database Schema

Complete Table Definitions • Relationships • Primary & Foreign Keys

Stack: React (Frontend) + Laravel (Backend) + MySQL

Revision Note — Changes applied per engineering review:

- contractor_master added as distinct entity (separated from vendor_master)
- stock_master added as single unified stock table (product × warehouse)
- financial_transaction_ledger (4.7) added — all voucher types linked here
- 3.2 production_routing removed; man/machine hours merged into production_bom
- 3.9 external_iqc added; 1.12 / 2.10 / 6.2 / 8.5 connected to financial ledger
- 2.7 purchase_iqc linked to qc_iqc_incoming (Module 7)
- 3.10 job_work_billbook linked to contractor_salary_sheet (Module 8)
- 5.3 salary_structure linked to 5.2 salary_head_master
- 6.3 logistics_challan_out added
- Module 12 statutory tables linked to all relevant source modules

How to Read This Document

Symbol / Colour	Meaning
PK (gold bg)	Primary Key — uniquely identifies each row
FK (green bg)	Foreign Key — links to another table
NOT NULL	Column is mandatory; cannot be empty
UNIQUE	All values must be distinct across rows
DEFAULT ...	Value auto-assigned when none is provided
ENUM(...)	Column only accepts one of the listed values
NEW	Table or column added in this revision
CHANGED	Table or column modified in this revision

Total Tables: 108 | Total Relationships: 97 | Modules: 13 + Master + Auth

Table Relationships (Foreign Key Map)

Child Table	Foreign Key	Parent Table	Parent Key	Relationship
users	role_id	roles	id	Many → One
permissions	role_id	roles	id	Many → One
inquiries	customer_id	customer_master	customer_id	Many → One
inquiries	sales_person_id	users	id	Many → One
inquiry_items	inquiry_no	inquiries	inquiry_no	Many → One
inquiry_items	product_id	product_master	product_id	Many → One
stock_master	product_id	product_master	product_id	Many → One
stock_master	warehouse_id	warehouse_master	warehouse_id	Many → One
quotations	inquiry_no	inquiries	inquiry_no	Many → One
quotation_items	quote_id	quotations	quote_id	Many → One
quotation_items	product_id	product_master	product_id	Many → One
customer_pos	customer_id	customer_master	customer_id	Many → One
sale_orders	po_number	customer_pos	po_number	One → One

sale_orders	customer_id	customer_master	customer_id	Many → One
dispatch_advice	so_no	sale_orders	so_no	Many → One
dispatch_advice	transporter_id	transport_master	transporter_id	Many → One
invoices	so_no	sale_orders	so_no	One → One
invoices	customer_id	customer_master	customer_id	Many → One
communication_log	invoice_no	invoices	invoice_no	Many → One
voucher_receipts	customer_id	customer_master	customer_id	Many → One
voucher_receipts	invoice_no	invoices	invoice_no	Many → One
voucher_receipts	finance_ledger_id	financial_transaction_ledger	ledger_id	Many → One
material_indents	dept_id	department_master	dept_id	Many → One
material_indent_items	indent_no	material_indents	indent_no	Many → One
material_indent_items	product_id	product_master	product_id	Many → One
purchase_orders	vendor_id	vendor_master	vendor_id	Many → One
purchase_order_items	po_no	purchase_orders	po_no	Many → One
grn	po_no	purchase_orders	po_no	Many → One
grn	vendor_id	vendor_master	vendor_id	Many → One
purchase_iqc	grn_no	grn	grn_no	Many → One
purchase_iqc	qc_ref_id	qc_iqc_incoming	id	One → One
material_receipts	grn_no	grn	grn_no	One → One
material_receipts	warehouse_id	warehouse_master	warehouse_id	Many → One
purchase_billbook	grn_no	grn	grn_no	One → One
purchase_billbook	vendor_id	vendor_master	vendor_id	Many → One
voucher_payments	vendor_id	vendor_master	vendor_id	Many → One
voucher_payments	finance_ledger_id	financial_transaction_ledger	ledger_id	Many → One
production_bom	finished_good_id	product_master	product_id	Many → One
production_bom	raw_material_id	product_master	product_id	Many → One
production_routecards	product_id	product_master	product_id	Many → One
material_issues	route_card_no	production_routecards	route_card_no	Many → One
production_reports	route_card_no	production_routecards	route_card_no	Many → One
job_orders	contractor_id	contractor_master	contractor_id	Many → One
challan_out	job_order_no	job_orders	job_order_no	Many → One
external_grn	challan_no	challan_out	challan_no	One → One
external_iqc	egrn_no	external_grn	egrn_no	One → One
external_iqc	product_id	product_master	product_id	Many → One
job_work_billbook	job_order_no	job_orders	job_order_no	Many → One
job_work_billbook	contractor_id	contractor_master	contractor_id	Many → One
job_work_billbook	salary_sheet_id	contractor_salary_sheet	id	Many → One
routecard_closures	route_card_no	production_routecards	route_card_no	One → One
financial_transaction_ledger	created_by	users	id	Many → One

voucher_journal	created_by	users	id	Many → One
voucher_journal	ledger_id	financial_transaction_ledger	ledger_id	One → One
voucher_payment_receipt	created_by	users	id	Many → One
voucher_payment_receipt	ledger_id	financial_transaction_ledger	ledger_id	One → One
voucher_contra	created_by	users	id	Many → One
voucher_contra	ledger_id	financial_transaction_ledger	ledger_id	One → One
journal_voucher_gst	created_by	users	id	Many → One
journal_voucher_gst	ledger_id	financial_transaction_ledger	ledger_id	One → One
employee_master	dept_id	department_master	dept_id	Many → One
employee_salary_structure	emp_code	employee_master	emp_code	Many → One
employee_salary_structure	salary_head_id	salary_head_master	id	Many → One
employee_salary_sheet	emp_code	employee_master	emp_code	Many → One
employee_salary_sheet	ledger_id	financial_transaction_ledger	ledger_id	Many → One
employee_advance_memo	emp_code	employee_master	emp_code	Many → One
transport_orders	transporter_id	transport_master	transporter_id	Many → One
logistics_challan_out	transport_order_no	transport_orders	order_no	Many → One
logistics_challan_out_items	challan_no	logistics_challan_out	challan_no	Many → One
logistics_challan_out_items	product_id	product_master	product_id	Many → One
freight_billbook	transport_order_no	transport_orders	order_no	Many → One
freight_billbook	transporter_id	transport_master	transporter_id	Many → One
freight_billbook	finance_ledger_id	financial_transaction_ledger	ledger_id	Many → One
qc_iqc_incoming	grn_no	grn	grn_no	Many → One
qc_mts	mta_no	mta	mta_no	Many → One
qc_pqc	route_card_no	production_routecards	route_card_no	Many → One
qc_pdi	so_no	sale_orders	so_no	Many → One
contractor_employee_master	contractor_id	contractor_master	contractor_id	Many → One
contractor_salary_structure	worker_id	contractor_employee_master	worker_id	Many → One
contractor_salary_structure	salary_head_id	contractor_salary_head_master	id	Many → One
contractor_salary_sheet	worker_id	contractor_employee_master	worker_id	Many → One

contractor_salary_sheet	contractor_id	contractor_master	contractor_id	Many → One
contractor_advance_memo	contractor_id	contractor_master	contractor_id	Many → One
contractor_voucher_payment	contractor_id	contractor_master	contractor_id	Many → One
contractor_voucher_payment	finance_ledger_id	financial_transaction_ledger	ledger_id	Many → One
tool_maintenance_chart	asset_code	tool_master	asset_code	Many → One
tool_calibration_report	asset_code	tool_master	asset_code	Many → One
tool_maintenance_memo	asset_code	tool_master	asset_code	Many → One
warehouse_opening	warehouse_id	warehouse_master	warehouse_id	Many → One
stock_transfers	from_warehouse_id	warehouse_master	warehouse_id	Many → One
stock_transfers	to_warehouse_id	warehouse_master	warehouse_id	Many → One
asset_addition_memo	asset_tag	fix_asset_master	asset_tag	One → One
asset_allocation_master	asset_tag	fix_asset_master	asset_tag	Many → One
asset_allocation_master	emp_code	employee_master	emp_code	Many → One
asset_sale_memo	asset_tag	fix_asset_master	asset_tag	Many → One
asset_depreciation_voucher	asset_tag	fix_asset_master	asset_tag	Many → One
asset_depreciation_voucher	ledger_id	financial_transaction_ledger	ledger_id	One → One
gstr1_upload	invoice_no	invoices	invoice_no	One → One
gst2a_reconciliation	vendor_id	vendor_master	vendor_id	Many → One
gst2a_reconciliation	bill_id	purchase_billbook	bill_id	Many → One
gst_deposit_challan	journal_voucher_gst_id	journal_voucher_gst	id	Many → One
gst_deposit_challan	ledger_id	financial_transaction_ledger	ledger_id	Many → One
tds_trace_details	voucher_payment_no	voucher_payments	voucher_no	Many → One
tds_trace_details	contractor_payment_no	contractor_voucher_payment	voucher_no	Many → One
tcs_details	invoice_no	invoices	invoice_no	One → One
balance_sheet	as_on_date	financial_transaction_ledger	entry_date	Many → One
mps_items	mps_id	master_production_schedule	mps_id	Many → One
mps_items	product_id	product_master	product_id	Many → One
mrp_output	mps_id	master_production_schedule	mps_id	Many → One
crp_output	mps_id	master_production_schedule	mps_id	One → One

cost_estimation	mps_id	master_production_sche dule	mps_id	One → One
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Module 0. System / Authentication

0.1 users

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	System login user
name	VARCHAR(150)	NOT NULL	---	Full name of user
email	VARCHAR(150)	NOT NULL, UNIQUE	---	Login email
password	VARCHAR(255)	NOT NULL	---	Hashed password (bcrypt)
FK role_id	BIGINT UNSIGNED	NOT NULL	FK → roles.id	Assigned role
is_active	TINYINT(1)	DEFAULT 1	---	1=Active, 0=Disabled
last_login_at	TIMESTAMP	NULLABLE	---	Last login timestamp
created_at	TIMESTAMP	DEFAULT NOW()	---	Record creation time
updated_at	TIMESTAMP	DEFAULT NOW()	---	Record last update

0.2 roles

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Role identifier
role_name	VARCHAR(100)	NOT NULL, UNIQUE	---	e.g. Super Admin, Sales Person
description	TEXT	NULLABLE	---	Role description

0.3 permissions

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Permission record
FK role_id	BIGINT UNSIGNED	NOT NULL	FK → roles.id	Role this permission belongs to
module	VARCHAR(100)	NOT NULL	---	Module name e.g. Sales, Purchase
page	VARCHAR(150)	NOT NULL	---	Specific page/sub-module

can_view	TINYINT(1)	DEFAULT 0	---	View permission flag
can_create	TINYINT(1)	DEFAULT 0	---	Create permission flag
can_edit	TINYINT(1)	DEFAULT 0	---	Edit permission flag
can_delete	TINYINT(1)	DEFAULT 0	---	Delete permission flag

Module: Master Tables

M.1 customer_master

Column Name	Data Type	Constraint	References	Description
PK customer_id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Unique customer identifier
customer_name	VARCHAR(200)	NOT NULL	---	Legal/trade name
gstn	VARCHAR(20)	NULLABLE, UNIQUE	---	GST Identification Number
pan	VARCHAR(15)	NULLABLE	---	PAN card number
email	VARCHAR(150)	NULLABLE	---	Primary email
phone	VARCHAR(20)	NOT NULL	---	Primary phone/WhatsApp
billing_address	TEXT	NULLABLE	---	Default billing address
shipping_addresses	TEXT	NULLABLE	---	Default shipping address
credit_period_days	SMALLINT	DEFAULT 30	---	Payment credit days
contact_person	VARCHAR(150)	NULLABLE	---	Primary contact name
state	VARCHAR(100)	NULLABLE	---	State for GST purposes
is_active	TINYINT(1)	DEFAULT 1	---	Active/Inactive flag

M.2 vendor_master

Stores suppliers for raw material purchase only. For job work / outsourcing, see contractor_master (M.4).

Column Name	Data Type	Constraint	References	Description
PK vendor_id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Unique vendor identifier
vendor_name	VARCHAR(200)	NOT NULL	---	Vendor company name
gstn	VARCHAR(20)	NULLABLE, UNIQUE	---	GST Identification Number
pan	VARCHAR(15)	NULLABLE	---	PAN card number

email	VARCHAR(150)	NULLABLE	---	Email for communication
phone	VARCHAR(20)	NOT NULL	---	Contact number
address	TEXT	NULLABLE	---	Registered address
bank_account_no	VARCHAR(30)	NULLABLE	---	Bank account number
bank_ifsc	VARCHAR(15)	NULLABLE	---	Bank IFSC code
payment_terms	VARCHAR(200)	NULLABLE	---	Default payment terms
is_active	TINYINT(1)	DEFAULT 1	---	Active/Inactive flag

M.3 product_master

Column Name	Data Type	Constraint	References	Description
PK product_id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Unique product identifier
product_name	VARCHAR(200)	NOT NULL	---	Product description
product_code	VARCHAR(50)	NOT NULL, UNIQUE	---	Internal item code
hsn_code	VARCHAR(20)	NULLABLE	---	HSN/SAC code for GST
unit_of_measure	VARCHAR(20)	NOT NULL	---	e.g. Pcs, Kg, Mtr
barcode	VARCHAR(50)	NULLABLE, UNIQUE	---	Barcode / QR code value
product_type	ENUM(...)	NOT NULL	---	Raw Material / Finished Good / Semi-Finished
reorder_level	DECIMAL(10, 2)	DEFAULT 0	---	Minimum stock trigger level
last_purchase_price	DECIMAL(12, 2)	NULLABLE	---	Latest purchase price for costing
is_active	TINYINT(1)	DEFAULT 1	---	Active/Inactive flag

M.4 contractor_master

NEW: Separate entity from vendor_master. Used exclusively for job work / outsourced labour contracts. All contractor_* tables and job_orders now reference contractor_id here instead of vendor_master.

Column Name	Data Type	Constraint	References	Description
PK contractor_id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Unique contractor identifier
contractor_name	VARCHAR(200)	NOT NULL	---	Contractor firm or individual name
gstn	VARCHAR(20)	NULLABLE	---	GST Identification Number
pan	VARCHAR(15)	NULLABLE	---	PAN card number

contact_person	VARCHAR(150)	NOT NULL	---	Primary contact name
mobile	VARCHAR(20)	NOT NULL	---	Contact mobile number
email	VARCHAR(150)	NULLABLE	---	Email address
address	TEXT	NULLABLE	---	Registered address
tds_section	VARCHAR(20)	NULLABLE	---	Applicable TDS section e.g. 194C
tds_rate	DECIMAL(5,2)	DEFAULT 0	---	TDS rate percentage
bank_account_no	VARCHAR(30)	NULLABLE	---	Bank account for payment
bank_ifsc	VARCHAR(15)	NULLABLE	---	Bank IFSC code
is_active	TINYINT(1)	DEFAULT 1	---	Active/Inactive flag

M.5 stock_master

NEW: Single unified stock table. Records current stock and blocked stock per product per warehouse. Replaces the ad-hoc current_stock fields scattered in module-level tables.

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Stock record identifier
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Product reference
FK warehouse_id	BIGINT UNSIGNED	NOT NULL	FK → warehouse_master.warehouse_id	Warehouse / location reference
current_stock	DECIMAL(12,4)	NOT NULL, DEFAULT 0	---	Physical stock on hand
blocked_stock	DECIMAL(12,4)	DEFAULT 0	---	Reserved for confirmed sale orders
net_available	DECIMAL(12,4)	GENERATED ALWAYS	---	current_stock – blocked_stock
last_updated	TIMESTAMP	DEFAULT NOW() ON UPDATE NOW()	---	Auto-updated on every transaction
UNIQUE	(product_id, warehouse_id)	UNIQUE	---	One row per product per warehouse

M.6 department_master

Column Name	Data Type	Constraint	References	Description
PK dept_id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Department identifier
dept_name	VARCHAR(150)	NOT NULL, UNIQUE	---	Department name
dept_head	VARCHAR(15)	NULLABLE	---	Department head name

	0)			
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M.7 transport_master

Column Name	Data Type	Constraint	References	Description
PK transporter_id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Transporter identifier
transporter_name	VARCHAR(200)	NOT NULL	---	Transport company name
owner_name	VARCHAR(150)	NOT NULL	---	Owner / contact name
mobile	VARCHAR(20)	NOT NULL	---	Contact mobile
gstn	VARCHAR(20)	NULLABLE	---	Transporter GSTIN

Module 1. Sales Management

1.1 inquiries

Column Name	Data Type	Constraint	References	Description
PK inquiry_no	VARCHAR(30)	PK, NOT NULL	---	Auto-generated inquiry number
FK customer_id	BIGINT UNSIGNED	NOT NULL	FK → customer_master.customer_id	Customer reference
FK sales_person_id	BIGINT UNSIGNED	NOT NULL	FK → users.id	Sales person assigned
inquiry_date	DATE	NOT NULL	---	Date inquiry was received
status	ENUM(...)	DEFAULT "New"	---	New / Processing / Quoted / Lost

1.2 inquiry_items

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Line item identifier
FK inquiry_no	VARCHAR(30)	NOT NULL	FK → inquiries.inquiry_no	Parent inquiry
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Product reference
quantity	DECIMAL(10, 2)	NOT NULL	---	Inquired quantity
target_price	DECIMAL(12, 2)	NULLABLE	---	Customer target price

1.3 stock_check (inquiry-time snapshot)

This table captures a point-in-time snapshot for delivery date calculation during inquiry processing. Live stock is maintained in stock_master (M.5).

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Stock check record
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Product checked
FK inquiry_no	VARCHAR(30)	NULLABLE	FK → inquiries.inquiry_no	Linked inquiry
current_stock	DECIMAL(10, 2)	NOT NULL	---	Snapshot of physical stock
blocked_stock	DECIMAL(10, 2)	DEFAULT 0	---	Reserved for pending orders
net_available	DECIMAL(10, 2)	GENERATED ALWAYS	---	current_stock – blocked_stock
delivery_date	DATE	NULLABLE	---	Auto-calculated by system logic
checked_at	TIMESTAMP	DEFAULT NOW()	---	When check was performed

1.4 quotations

Column Name	Data Type	Constraint	References	Description
PK quote_id	VARCHAR(30)	PK, NOT NULL	---	Auto-generated quotation ID
FK inquiry_no	VARCHAR(30)	NOT NULL	FK → inquiries.inquiry_no	Source inquiry
FK customer_id	BIGINT UNSIGNED	NOT NULL	FK → customer_master.customer_id	Customer
valid_until	DATE	NOT NULL	---	Quotation expiry date
payment_terms	VARCHAR(200)	NULLABLE	---	Payment terms text
created_at	TIMESTAMP	DEFAULT NOW()	---	Quotation creation date

1.5 quotation_items

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Line item identifier
FK quote_id	VARCHAR(30)	NOT NULL	FK → quotations.quote_id	Parent quotation
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Product

qty	DECIMAL(10, 2)	NOT NULL	---	Quoted quantity
rate	DECIMAL(12, 2)	NOT NULL	---	Quoted rate per unit
gst_percent	DECIMAL(5,2)	NOT NULL	---	Applicable GST percentage
total	DECIMAL(14, 2)	GENERATED ALWAYS	---	$qty \times rate \times (1 + \text{gst}/100)$

1.6 customer_pos

Column Name	Data Type	Constraint	References	Description
PK po_number	VARCHAR(50)	PK, NOT NULL	---	Customer PO number
FK quote_id	VARCHAR(30)	NULLABLE	FK → quotations.quote_id	Linked quotation
FK customer_id	BIGINT UNSIGNED	NOT NULL	FK → customer_master.customer_id	Customer
po_date	DATE	NOT NULL	---	PO issue date
po_file_path	VARCHAR(500)	NULLABLE	---	Uploaded PO document path
confirmed_delivery_date	DATE	NOT NULL	---	Agreed delivery date

1.7 sale_orders

Column Name	Data Type	Constraint	References	Description
PK so_no	VARCHAR(30)	PK, NOT NULL	---	Auto-generated SO number
FK po_number	VARCHAR(50)	NOT NULL	FK → customer_pos.po_number	Customer PO reference
FK customer_id	BIGINT UNSIGNED	NOT NULL	FK → customer_master.customer_id	Customer
billing_address	TEXT	NOT NULL	---	Invoice billing address
shipping_addresses	TEXT	NOT NULL	---	Delivery / shipping address
status	ENUM(...)	DEFAULT "Pending"	---	Pending / Dispatched / Closed
created_at	TIMESTAMP	DEFAULT NOW()	---	SO creation timestamp

1.8 dispatch_advice

Column Name	Data Type	Constraint	References	Description
PK dispatch_id	VARCHAR(30)	PK, NOT NULL	---	Dispatch advice identifier
FK so_no	VARCHAR(30)	NOT NULL	FK →	Sale order reference

	)		sale_orders.so_no	
FK transporter_id	BIGINT UNSIGNED	NOT NULL	FK → transport_master.transporter_id	Transporter
vehicle_no	VARCHAR(30)	NOT NULL	---	Vehicle registration number
driver_name	VARCHAR(150)	NOT NULL	---	Driver full name
dispatch_date	DATE	NOT NULL	---	Scheduled dispatch date

1.9 invoices

Column Name	Data Type	Constraint	References	Description
PK invoice_no	VARCHAR(30)	PK, NOT NULL	---	Auto-generated invoice number
FK so_no	VARCHAR(30)	NOT NULL	FK → sale_orders.so_no	Source sale order
FK customer_id	BIGINT UNSIGNED	NOT NULL	FK → customer_master.customer_id	Customer
invoice_date	DATE	NOT NULL	---	Date of invoice
place_of_supply	VARCHAR(100)	NOT NULL	---	State/place of supply for GST
e_way_bill_no	VARCHAR(50)	NULLABLE	---	E-Way Bill number
taxable_value	DECIMAL(14,2)	NOT NULL	---	Value before tax
gst_amount	DECIMAL(14,2)	NOT NULL	---	Total GST charged
round_off	DECIMAL(6,2)	DEFAULT 0	---	Rounding adjustment
grand_total	DECIMAL(14,2)	NOT NULL	---	Final payable amount
due_date	DATE	GENERATED ALWAYS	---	invoice_date + credit_period_days from customer_master

1.10 collection_reminder_config

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Config record ID
FK invoice_no	VARCHAR(30)	NOT NULL	FK → invoices.invoice_no	Invoice being tracked
FK customer_id	BIGINT UNSIGNED	NOT NULL	FK → customer_master.customer_id	Customer
credit_period	SMALLINT	NOT NULL	---	Credit days (fetched from customer master)
due_date	DATE	NOT NULL	---	Auto-calculated due date

reminder_setting	ENUM(...)	DEFAULT "Moderate"	---	Strict / Moderate / Lenient
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1.11 communication_log

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Communication record
FK invoice_no	VARCHAR(30)	NOT NULL	FK → invoices.invoice_no	Related invoice
FK customer_id	BIGINT UNSIGNED	NOT NULL	FK → customer_master.customer_id	Recipient customer
sent_date	TIMESTAMP	NOT NULL	---	Date and time message was sent
channel	ENUM(...)	NOT NULL	---	WhatsApp / Email
message_type	VARCHAR(100)	NOT NULL	---	Trigger type e.g. 7-day reminder
status	ENUM(...)	NOT NULL	---	Sent / Delivered / Read / Failed

1.12 voucher_receipts

CHANGED: Added finance_ledger_id FK → financial_transaction_ledger. Every payment receipt auto-creates a ledger entry (Debit: Bank/Cash, Credit: Customer/AR). This links the Sales module cash collection directly to the Finance books.

Column Name	Data Type	Constraint	References	Description
PK receipt_no	VARCHAR(30)	PK, NOT NULL	---	Auto-generated receipt number
FK customer_id	BIGINT UNSIGNED	NOT NULL	FK → customer_master.customer_id	Customer who paid
FK invoice_no	VARCHAR(30)	NULLABLE	FK → invoices.invoice_no	Invoice settled (partial allowed)
receipt_date	DATE	NOT NULL	---	Date payment received
amount	DECIMAL(14, 2)	NOT NULL	---	Amount received
mode	ENUM(...)	NOT NULL	---	Cheque / NEFT / UPI / Cash
ref_no	VARCHAR(100)	NULLABLE	---	Cheque / transaction reference number
FK finance_ledger_id	BIGINT UNSIGNED	NOT NULL	FK → financial_transaction_ledger.ledger_id	Auto-created double-entry in Finance ledger

Module 2. Purchase Management

2.1 material_indents

Column Name	Data Type	Constraint	References	Description
PK indent_no	VARCHAR(30)	PK, NOT NULL	---	Auto-generated indent number
FK dept_id	BIGINT UNSIGNED	NOT NULL	FK → department_master.dept_id	Requesting department
FK requested_by	BIGINT UNSIGNED	NOT NULL	FK → users.id	User who raised indent
request_date	DATE	NOT NULL	---	Date of indent request
priority	ENUM(...)	DEFAULT "Medium"	---	High / Medium / Low
status	ENUM(...)	DEFAULT "Open"	---	Open / PO Raised / Closed

2.2 material_indent_items

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Line item identifier
FK indent_no	VARCHAR(30)	NOT NULL	FK → material_indents.indent_no	Parent indent
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Item requested
current_stock	DECIMAL(10, 2)	NOT NULL	---	Stock snapshot at time of request
requested_qty	DECIMAL(10, 2)	NOT NULL	---	Quantity requested

2.3 purchase_orders

Column Name	Data Type	Constraint	References	Description
PK po_no	VARCHAR(30)	PK, NOT NULL	---	Auto-generated PO number
FK vendor_id	BIGINT UNSIGNED	NOT NULL	FK → vendor_master.vendor_id	Vendor reference
FK indent_no	VARCHAR(30)	NULLABLE	FK → material_indents.indent_no	Source indent
po_date	DATE	NOT NULL	---	PO issue date
valid_until	DATE	NULLABLE	---	PO validity date
status	ENUM(...)	DEFAULT "Open"	---	Open / Partial / Closed

2.4 purchase_order_items

Column Name	Data Type	Constraint	References	Description
-------------	-----------	------------	------------	-------------

PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Line item identifier
FK po_no	VARCHAR(30)	NOT NULL	FK → purchase_orders.po_no	Parent PO
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Product ordered
qty	DECIMAL(10, 2)	NOT NULL	---	Ordered quantity
rate	DECIMAL(12, 2)	NOT NULL	---	Agreed rate per unit
expected_delivery_date	DATE	NOT NULL	---	Expected delivery date

2.5 purchase_schedules

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Schedule record ID
FK po_no	VARCHAR(30)	NOT NULL	FK → purchase_orders.po_no	Parent PO
expected_date	DATE	NOT NULL	---	Expected delivery date
follow_up_status	ENUM(...)	DEFAULT "On-Time"	---	On-Time / Delayed
remarks	TEXT	NULLABLE	---	Follow-up notes

2.6 grn (Goods Receipt Note)

Column Name	Data Type	Constraint	References	Description
PK grn_no	VARCHAR(30)	PK, NOT NULL	---	Goods Receipt Note number
FK po_no	VARCHAR(30)	NOT NULL	FK → purchase_orders.po_no	Source PO
FK vendor_id	BIGINT UNSIGNED	NOT NULL	FK → vendor_master.vendor_id	Vendor
vendor_challan_no	VARCHAR(50)	NOT NULL	---	Vendor delivery challan number
gate_entry_date	DATE	NOT NULL	---	Date material arrived at gate
vehicle_no	VARCHAR(30)	NULLABLE	---	Vehicle that delivered goods

2.7 purchase_iqc

CHANGED: Added qc_ref_id FK → qc_iqc_incoming (Module 7). Purchase IQC result is now formally cross-referenced with the Quality module's IQC table so QC team can view/act from both modules without data duplication.

Column Name	Data Type	Constraint	References	Description
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PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	IQC record identifier
FK grn_no	VARCHAR(30)	NOT NULL	FK → grn.grn_no	Source GRN
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Item inspected
total_qty	DECIMAL(10, 2)	NOT NULL	---	Total received quantity
sample_size	DECIMAL(10, 2)	NOT NULL	---	Sample inspected
accepted_qty	DECIMAL(10, 2)	NOT NULL	---	Quantity accepted
rejected_qty	DECIMAL(10, 2)	NOT NULL	---	Quantity rejected
result	ENUM(...)	NOT NULL	---	Pass / Fail / Partial
FK qc_ref_id	BIGINT UNSIGNED	NULLABLE	FK → qc_iqc_incoming.id	Cross-reference to Quality module IQC record

2.8 material_receipts

Column Name	Data Type	Constraint	References	Description
PK receipt_id	VARCHAR(30)	PK, NOT NULL	---	Material receipt identifier
FK grn_no	VARCHAR(30)	NOT NULL	FK → grn.grn_no	Source GRN
FK warehouse_id	BIGINT UNSIGNED	NOT NULL	FK → warehouse_master.warehouse_id	Storage location
storage_location	VARCHAR(100)	NOT NULL	---	Rack/Bin detail
batch_no	VARCHAR(50)	NULLABLE	---	Batch/lot number
received_date	DATE	NOT NULL	---	Date added to stock

2.9 purchase_billbook

Column Name	Data Type	Constraint	References	Description
PK bill_id	VARCHAR(30)	PK, NOT NULL	---	Internal bill identifier
FK grn_no	VARCHAR(30)	NOT NULL	FK → grn.grn_no	Source GRN
FK vendor_id	BIGINT UNSIGNED	NOT NULL	FK → vendor_master.vendor_id	Vendor
vendor_invoice_no	VARCHAR(50)	NOT NULL	---	Vendor invoice number
invoice_date	DATE	NOT NULL	---	Vendor invoice date
taxable_value	DECIMAL(14,	NOT NULL	---	Value before tax

	2)			
gst_amount	DECIMAL(14, 2)	NOT NULL	---	GST charged by vendor
total	DECIMAL(14, 2)	NOT NULL	---	Total payable amount

2.10 voucher_payments (Purchase)

CHANGED: Added finance_ledger_id FK → financial_transaction_ledger. Every vendor payment auto-creates a ledger entry (Debit: Accounts Payable/Vendor, Credit: Bank). This links Purchase payments directly to Finance books.

Column Name	Data Type	Constraint	References	Description
PK voucher_no	VARCHAR(30)	PK, NOT NULL	---	Auto-generated voucher number
FK vendor_id	BIGINT UNSIGNED	NOT NULL	FK → vendor_master.vendor_id	Vendor paid
FK bill_id	VARCHAR(30)	NULLABLE	FK → purchase_billbook.bill_id	Bill settled
payment_date	DATE	NOT NULL	---	Date of payment
amount	DECIMAL(14, 2)	NOT NULL	---	Amount paid
bank_account	VARCHAR(100)	NOT NULL	---	Company bank account used
mode	ENUM(...)	NOT NULL	---	NEFT / RTGS / Cheque / Cash
FK finance_ledger_id	BIGINT UNSIGNED	NOT NULL	FK → financial_transaction_ledger.ledger_id	Auto-created double-entry in Finance ledger

Module 3. Production Management

3.1 production_bom

CHANGED: Added man_hours_per_unit, machine_hours_per_unit, and sequence columns (migrated from the removed production_routing table) so BOM holds both material and time-standard data in one place.

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	BOM record identifier
FK finished_good_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Finished product
FK raw_material_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Raw material used
process_name	VARCHAR(150)	NOT NULL	---	Process e.g. Cutting, Welding
machine	VARCHAR(100)	NULLABLE	---	Machine used for process

bom_qty_per_unit	DECIMAL(10,4)	NOT NULL	---	RM qty needed per 1 unit of FG
output_qty	DECIMAL(10,4)	NOT NULL	---	Expected FG output quantity
man_hours_per_unit	DECIMAL(8,4)	NOT NULL, DEFAULT 0	---	Man-hours required per unit (from routing)
machine_hours_per_unit	DECIMAL(8,4)	NOT NULL, DEFAULT 0	---	Machine hours per unit (from routing)
sequence	SMALLINT	NOT NULL, DEFAULT 1	---	Process sequence order

Table 3.2 production_routing has been REMOVED. Its man_hours_per_unit, machine_hours_per_unit, and sequence columns are now part of production_bom above. This eliminates the redundant join and keeps all BOM + routing data co-located.

3.2 production_routecards

Column Name	Data Type	Constraint	References	Description
PK route_card_no	VARCHAR(30)	PK, NOT NULL	---	Route card number
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Product to produce
batch_no	VARCHAR(50)	NOT NULL	---	Batch/lot number
plan_qty	DECIMAL(10,2)	NOT NULL	---	Planned production quantity
start_date	DATE	NOT NULL	---	Planned start date
end_date	DATE	NOT NULL	---	Planned completion date
status	ENUM(...)	DEFAULT "Open"	---	Open / In Progress / Closed

3.3 material_issues

Column Name	Data Type	Constraint	References	Description
PK issue_id	VARCHAR(30)	PK, NOT NULL	---	Material issue identifier
FK route_card_no	VARCHAR(30)	NOT NULL	FK → production_routecards.route_card_no	Production batch
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Item issued
qty_requested	DECIMAL(10,2)	NOT NULL	---	Quantity requested from store
qty_issued	DECIMAL(10,2)	NOT NULL	---	Actual quantity issued
issued_date	DATE	NOT NULL	---	Date of issue

3.4 mta (Material Transfer Acknowledgement)

Column Name	Data Type	Constraint	References	Description
PK mta_no	VARCHAR(30)	PK, NOT NULL	---	MTA number
FK from_dept_id	BIGINT UNSIGNED	NOT NULL	FK → department_master.dept_id	Source department
FK to_dept_id	BIGINT UNSIGNED	NOT NULL	FK → department_master.dept_id	Destination department
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Item transferred
qty	DECIMAL(10, 2)	NOT NULL	---	Quantity transferred
received_by	VARCHAR(150)	NOT NULL	---	Person who received material
transfer_date	DATE	NOT NULL	---	Date of transfer

3.5 production_reports

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Production report record
FK route_card_no	VARCHAR(30)	NULLABLE	FK → production_routes.route_card_no	Linked batch
report_date	DATE	NOT NULL	---	Production date
shift	ENUM(...)	NOT NULL	---	Morning / Evening / Night
machine_no	VARCHAR(50)	NOT NULL	---	Machine identifier
operator	VARCHAR(150)	NOT NULL	---	Operator name
production_qty	DECIMAL(10, 2)	NOT NULL	---	Total units produced
rejection_qty	DECIMAL(10, 2)	DEFAULT 0	---	Units rejected

3.6 job_orders

CHANGED: contractor_id FK now references contractor_master.contractor_id instead of vendor_master. Vendors (raw material suppliers) and Contractors (job work firms) are distinct entities.

Column Name	Data Type	Constraint	References	Description
PK job_order_no	VARCHAR(30)	PK, NOT NULL	---	External job order number
FK contractor_id	BIGINT UNSIGNED	NOT NULL	FK → contractor_master.contractor_id	Contractor for job work
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Item sent out

process_required	VARCHAR(200)	NOT NULL	---	Process to be done externally
rate	DECIMAL(10,2)	NOT NULL	---	Rate per unit for job work
issue_date	DATE	NOT NULL	---	Date material was sent out
status	ENUM(...)	DEFAULT "Open"	---	Open / Received / Billed

3.7 challan_out (Production)

Column Name	Data Type	Constraint	References	Description
PK challan_no	VARCHAR(30)	PK, NOT NULL	---	Outward challan number
FK job_order_no	VARCHAR(30)	NULLABLE	FK → job_orders.job_order_no	Job order reference
FK so_no	VARCHAR(30)	NULLABLE	FK → sale_orders.so_no	Sale order reference
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Item dispatched
qty	DECIMAL(10,2)	NOT NULL	---	Quantity sent
vehicle_no	VARCHAR(30)	NULLABLE	---	Vehicle used
challan_date	DATE	NOT NULL	---	Date of dispatch

3.8 external_grn

Column Name	Data Type	Constraint	References	Description
PK egrn_no	VARCHAR(30)	PK, NOT NULL	---	External GRN number
FK challan_no	VARCHAR(30)	NOT NULL	FK → challan_out.challan_no	Source challan
received_qty	DECIMAL(10,2)	NOT NULL	---	Quantity received back
passed_qty	DECIMAL(10,2)	NOT NULL	---	Quantity that passed initial count
rejected_qty	DECIMAL(10,2)	DEFAULT 0	---	Quantity rejected on receipt
received_date	DATE	NOT NULL	---	Date received from contractor

3.9 external_iqc

NEW: New table. Performs quality inspection on goods returned by the contractor after external job work. Mirrors purchase_iqc logic for the external job-work cycle.

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	External IQC record ID

FK egrn_no	VARCHAR(30)	NOT NULL	FK → external_grn.egrn_no	Source external GRN (One-to-One)
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Item inspected
total_received_qty	DECIMAL(10, 2)	NOT NULL	---	Total qty received from contractor
sample_size	DECIMAL(10, 2)	NOT NULL	---	Sample size inspected
accepted_qty	DECIMAL(10, 2)	NOT NULL	---	Quantity accepted after IQC
rejected_qty	DECIMAL(10, 2)	DEFAULT 0	---	Quantity rejected after IQC
rejection_reason	TEXT	NULLABLE	---	Reason for rejection if any
result	ENUM(...)	NOT NULL	---	Pass / Fail / Partial
inspected_by	VARCHAR(150)	NOT NULL	---	QC inspector name
inspection_date	DATE	NOT NULL	---	Date of inspection

3.10 job_work_billbook

CHANGED: contractor_id now references contractor_master. Added salary_sheet_id FK → contractor_salary_sheet to link the job bill with the contractor's salary record — ensuring the contractor payment (Module 8) and the billbook are reconciled.

Column Name	Data Type	Constraint	References	Description
PK bill_no	VARCHAR(30)	PK, NOT NULL	---	Job work bill number
FK job_order_no	VARCHAR(30)	NOT NULL	FK → job_orders.job_order_no	Job order reference
FK contractor_id	BIGINT UNSIGNED	NOT NULL	FK → contractor_master.contractor_id	Contractor billed
FK salary_sheet_id	BIGINT UNSIGNED	NULLABLE	FK → contractor_salary_sheet.id	Linked contractor salary sheet record (Module 8)
labor_charges	DECIMAL(12, 2)	NOT NULL	---	Total labour charges
gst_percent	DECIMAL(5,2)	NOT NULL	---	GST rate applied
gst_amount	DECIMAL(12, 2)	NOT NULL	---	GST amount
total	DECIMAL(14, 2)	NOT NULL	---	Total payable

3.11 routecard_closures

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Closure record

FK route_card_no	VARCHAR(30)	NOT NULL	FK → production_routes.route_card_no	Batch closed
final_fg_qty	DECIMAL(10, 2)	NOT NULL	---	Final finished goods quantity
scrap_generated	DECIMAL(10, 2)	DEFAULT 0	---	Scrap quantity from batch
closure_date	DATE	NOT NULL	---	Date batch was closed

Module 4. Finance Management

Design principle: Every money movement (receipt, payment, journal, contra, GST adjustment, depreciation, salary disbursement, freight) is routed through financial_transaction_ledger (4.7) as the single source of truth. Module-specific voucher tables retain their operational detail but each carries a finance_ledger_id FK to their corresponding double-entry record.

4.1 voucher_journal

CHANGED: Added ledger_id FK → financial_transaction_ledger. Manual journal entries are now also reflected in the unified ledger.

Column Name	Data Type	Constraint	References	Description
PK journal_no	VARCHAR(30)	PK, NOT NULL	---	Journal entry number
FK created_by	BIGINT UNSIGNED	NOT NULL	FK → users.id	Accountant user
entry_date	DATE	NOT NULL	---	Date of journal entry
debit_account	VARCHAR(150)	NOT NULL	---	Debit ledger account name
credit_account	VARCHAR(150)	NOT NULL	---	Credit ledger account name
amount	DECIMAL(14, 2)	NOT NULL	---	Transaction amount
narration	TEXT	NULLABLE	---	Explanation / narration
FK ledger_id	BIGINT UNSIGNED	NOT NULL	FK → financial_transaction_ledger.ledger_id	Corresponding ledger entry

4.2 voucher_payment_receipt

CHANGED: Added ledger_id FK. Finance-module generic payment/receipt entries (not tied to a specific vendor or customer) are logged here and mirrored in the ledger.

Column Name	Data Type	Constraint	References	Description
PK voucher_no	VARCHAR(30)	PK, NOT NULL	---	Voucher number
voucher_type	ENUM(...)	NOT NULL	---	Payment / Receipt
FK created_by	BIGINT UNSIGNED	NOT NULL	FK → users.id	Accountant user
entry_date	DATE	NOT NULL	---	Transaction date

party_name	VARCHAR(200)	NOT NULL	---	Vendor or customer name
amount	DECIMAL(14, 2)	NOT NULL	---	Transaction amount
mode	ENUM(...)	NOT NULL	---	Cash / Bank / Cheque / NEFT
FK ledger_id	BIGINT UNSIGNED	NOT NULL	FK → financial_transaction_ledger.ledger_id	Corresponding ledger entry

4.3 voucher_contra

CHANGED: Added ledger_id FK. Contra entries (cash deposit / withdrawal) are mirrored in the ledger.

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Contra entry ID
entry_date	DATE	NOT NULL	---	Transaction date
from_account	VARCHAR(150)	NOT NULL	---	Source account e.g. Cash
to_account	VARCHAR(150)	NOT NULL	---	Destination account e.g. Bank
amount	DECIMAL(14, 2)	NOT NULL	---	Amount transferred
FK created_by	BIGINT UNSIGNED	NOT NULL	FK → users.id	User who created entry
FK ledger_id	BIGINT UNSIGNED	NOT NULL	FK → financial_transaction_ledger.ledger_id	Corresponding ledger entry

4.4 journal_voucher_gst

CHANGED: Added ledger_id FK. GST adjustments (ITC claims, reversals) are mirrored in the ledger and consumed by gst_deposit_challan in Module 12.

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	GST journal entry ID
entry_date	DATE	NOT NULL	---	Entry date
gst_ledger	ENUM(...)	NOT NULL	---	Input / Output
adjustment_type	VARCHAR(100)	NOT NULL	---	e.g. Reversal, ITC Claim
amount	DECIMAL(14, 2)	NOT NULL	---	Adjustment amount
FK created_by	BIGINT UNSIGNED	NOT NULL	FK → users.id	Accountant
FK ledger_id	BIGINT UNSIGNED	NOT NULL	FK → financial_transaction_ledger.ledger_id	Corresponding ledger entry

4.5 bank_reconciliation

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Reconciliation record
bank_account	VARCHAR(150)	NOT NULL	---	Bank account name/number
statement_date	DATE	NOT NULL	---	Bank statement date
system_balance	DECIMAL(16,2)	NOT NULL	---	Balance as per ERP books
bank_balance	DECIMAL(16,2)	NOT NULL	---	Balance as per bank statement
unreconciled_amount	DECIMAL(16,2)	GENERATED ALWAYS	---	system_balance – bank_balance

4.6 credit_card_statements

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Statement entry ID
card_no	VARCHAR(25)	NOT NULL	---	Last 4 digits of card
statement_month	VARCHAR(10)	NOT NULL	---	YYYY-MM period
transaction_date	DATE	NOT NULL	---	Individual transaction date
merchant	VARCHAR(200)	NOT NULL	---	Merchant name
amount	DECIMAL(12,2)	NOT NULL	---	Transaction amount
expense_head	VARCHAR(100)	NOT NULL	---	Expense category / GL head

4.7 financial_transaction_ledger

NEW: Central ledger table. All voucher types from all modules post a double-entry record here. This table is the single source of truth for P&L, Balance Sheet, and any financial report. Accountants can view consolidated books without querying individual module tables.

Column Name	Data Type	Constraint	References	Description
PK ledger_id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Unique ledger entry ID
transaction_type	ENUM(...)	NOT NULL	---	Receipt / Payment / Journal / Contra / GST_Adj / Depreciation / Salary / Freight / Contractor_Payment / Job_Work
source_module	VARCHAR(50)	NOT NULL	---	Origin module e.g. Sales, Purchase, HR, Logistics
source_ref	VARCHAR(50)	NOT NULL	---	Source document number (receipt_no / voucher_no / bill_no etc.)

party_type	ENUM(...)	NULLABLE	---	Customer / Vendor / Contractor / Employee / Internal
party_id	BIGINT UNSIGNED	NULLABLE	---	ID in the respective master table
party_name	VARCHAR(200)	NULLABLE	---	Denormalised party name for quick reporting
debit_account	VARCHAR(150)	NOT NULL	---	Debit GL account name
credit_account	VARCHAR(150)	NOT NULL	---	Credit GL account name
amount	DECIMAL(16, 2)	NOT NULL	---	Transaction amount
entry_date	DATE	NOT NULL	---	Effective accounting date
narration	TEXT	NULLABLE	---	Auto-populated or manual description
FK created_by	BIGINT UNSIGNED	NOT NULL	FK → users.id	User / system that created entry
created_at	TIMESTAMP	DEFAULT NOW()	---	System timestamp

Module 5. HR Management

5.1 employee_master

Column Name	Data Type	Constraint	References	Description
PK emp_code	VARCHAR(20)	PK, NOT NULL	---	Employee code
emp_name	VARCHAR(150)	NOT NULL	---	Full name
FK dept_id	BIGINT UNSIGNED	NOT NULL	FK → department_master.dept_id	Department assigned
designation	VARCHAR(100)	NOT NULL	---	Job title / designation
mobile	VARCHAR(20)	NOT NULL	---	Contact number
joining_date	DATE	NOT NULL	---	Date of joining
basic_salary	DECIMAL(12, 2)	NOT NULL	---	Basic monthly salary
bank_account_no	VARCHAR(30)	NULLABLE	---	Bank account for salary
bank_ifsc	VARCHAR(15)	NULLABLE	---	Bank IFSC code
is_active	TINYINT(1)	DEFAULT 1	---	Active/Inactive flag

5.2 salary_head_master

Column Name	Data Type	Constraint	References	Description
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PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Salary head identifier
head_name	VARCHAR(100)	NOT NULL, UNIQUE	---	e.g. HRA, PF, DA, Bonus
head_type	ENUM(...)	NOT NULL	---	Earning / Deduction
is_statutory	TINYINT(1)	DEFAULT 0	---	1 = statutory deduction (PF/ESI/TDS)

5.3 employee_salary_structure

CHANGED: Added salary_head_id FK → salary_head_master. Each structure row now explicitly references which salary component it defines, enabling dynamic payslip generation from component-level records.

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Structure record ID
FK emp_code	VARCHAR(20)	NOT NULL	FK → employee_master.emp_code	Employee
FK salary_head_id	BIGINT UNSIGNED	NOT NULL	FK → salary_head_master.id	Salary component being defined
effective_date	DATE	NOT NULL	---	Date structure becomes active
amount	DECIMAL(12,2)	NOT NULL	---	Value for this head (flat or percent-based)
is_percent	TINYINT(1)	DEFAULT 0	---	0=flat amount, 1=% of basic

5.4 employee_salary_sheet

CHANGED: Added ledger_id FK → financial_transaction_ledger. When salary is disbursed, a ledger entry is created (Debit: Salary Expense, Credit: Bank/Employee Payable), completing the Finance linkage.

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Payroll record ID
FK emp_code	VARCHAR(20)	NOT NULL	FK → employee_master.emp_code	Employee
month	TINYINT	NOT NULL	---	Payroll month (1-12)
year	YEAR	NOT NULL	---	Payroll year
total_days	TINYINT	NOT NULL	---	Total working days in month
present_days	DECIMAL(5,2)	NOT NULL	---	Days employee was present
calculated_gross	DECIMAL(14,2)	NOT NULL	---	Gross salary before deductions
deductions	DECIMAL(14,2)	DEFAULT 0	---	Total deductions (PF, TDS, advances etc.)
net_pay	DECIMAL(14,2)	GENERATED ALWAYS	---	calculated_gross – deductions
FK ledger_id	BIGINT UNSIGNED	NULLABLE	FK → financial_transaction_ledger	Finance ledger entry for this disbursement

			on_ledger.ledger_id	
--	--	--	---------------------	--

5.5 employee_advance_memo

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Advance record ID
FK emp_code	VARCHAR(20)	NOT NULL	FK → employee_master.emp_code	Employee
advance_date	DATE	NOT NULL	---	Date advance was given
amount	DECIMAL(12, 2)	NOT NULL	---	Advance amount
purpose	VARCHAR(300)	NULLABLE	---	Reason for advance
recovery_month	VARCHAR(10)	NULLABLE	---	YYYY-MM recovery month

Module 6. Logistics Management

6.1 transport_orders

Column Name	Data Type	Constraint	References	Description
PK order_no	VARCHAR(30)	PK, NOT NULL	---	Transport order number
FK transporter_id	BIGINT UNSIGNED	NOT NULL	FK → transport_master.transporter_id	Transporter
pickup_date	DATE	NOT NULL	---	Scheduled pickup date
destination	VARCHAR(300)	NOT NULL	---	Delivery destination
vehicle_type	VARCHAR(100)	NULLABLE	---	Type of vehicle booked
status	ENUM(...)	DEFAULT "Booked"	---	Booked / In Transit / Delivered

6.2 logistics_challan_out

NEW: Delivery document for the Logistics module, aligned with the workflow in the reference PDF (Section 6.3). Issued against a Transport Order and linked to the Sale Order being dispatched.

Column Name	Data Type	Constraint	References	Description
PK challan_no	VARCHAR(30)	PK, NOT NULL	---	Auto-generated logistics challan number
FK transport_order_no	VARCHAR(30)	NOT NULL	FK → transport_orders.order_no	Linked transport order
FK so_no	VARCHAR(30)	NULLABLE	FK → sale_orders.so_no	Sale order being dispatched

FK dispatch_id	VARCHAR(30)	NULLABLE	FK → dispatch_advice.dispatch_id	Dispatch advice reference
challan_date	DATE	NOT NULL	---	Date of challan issue
remarks	TEXT	NULLABLE	---	Additional delivery remarks

6.3 logistics_challan_out_items

NEW: Line items for the logistics challan (one row per product per challan).

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Line item ID
FK challan_no	VARCHAR(30)	NOT NULL	FK → logistics_challan_out.challan_no	Parent challan
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Item dispatched
qty	DECIMAL(10, 2)	NOT NULL	---	Quantity dispatched

6.4 freight_billbook

CHANGED: Added finance_ledger_id FK → financial_transaction_ledger. When a freight bill is booked, a ledger entry is created (Debit: Freight Expense, Credit: Transporter Payable), reflecting the liability in Finance books.

Column Name	Data Type	Constraint	References	Description
PK bill_no	VARCHAR(30)	PK, NOT NULL	---	Freight bill number
FK transport_order_no	VARCHAR(30)	NOT NULL	FK → transport_orders.order_no	Transport order
FK transporter_id	BIGINT UNSIGNED	NOT NULL	FK → transport_master.transporter_id	Transporter
bill_date	DATE	NOT NULL	---	Bill date
lr_no	VARCHAR(50)	NULLABLE	---	LR/GR number
freight_amount	DECIMAL(12, 2)	NOT NULL	---	Base freight amount
gst_percent	DECIMAL(5, 2)	NOT NULL	---	GST rate on freight
gst_amount	DECIMAL(12, 2)	NOT NULL	---	GST amount
total_payable	DECIMAL(14, 2)	NOT NULL	---	Total to be paid
FK finance_ledger_id	BIGINT UNSIGNED	NOT NULL	FK → financial_transaction_ledger.ledger_id	Auto-created Finance ledger entry

Module 7. Quality Management

7.1 qc_iqc_incoming

This is the Quality module's authoritative IQC record. purchase_iqc (2.7) cross-references this table via qc_ref_id, so QC team owns the inspection data while Purchase module tracks the GRN.

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	QC IQC record
FK grn_no	VARCHAR(30)	NOT NULL	FK → grn.grn_no	Source GRN
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Item inspected
sample_qty	DECIMAL(10, 2)	NOT NULL	---	Sample quantity inspected
visual_check	ENUM(...)	NOT NULL	---	Pass / Fail
dimension_check	ENUM(...)	NOT NULL	---	Pass / Fail
result	ENUM(...)	NOT NULL	---	Pass / Fail / Hold
inspected_by	VARCHAR(150)	NOT NULL	---	QC inspector name
inspection_date	DATE	NOT NULL	---	Date of inspection

7.2 qc_mts (Material Transfer Slip)

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	MTS record ID
FK mta_no	VARCHAR(30)	NOT NULL	FK → mta.mta_no	Material transfer reference
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Item checked
qty_checked	DECIMAL(10, 2)	NOT NULL	---	Quantity checked during move
status	ENUM(...)	NOT NULL	---	OK / Damaged
remarks	TEXT	NULLABLE	---	Observations if damaged

7.3 qc_pqc (Process Quality Control)

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	PQC record ID
FK route_card_no	VARCHAR(30)	NOT NULL	FK → production_routes.route_card_no	Production batch

stage_name	VARCHAR(150)	NOT NULL	---	Process stage checked
operator	VARCHAR(150)	NOT NULL	---	Operator at that stage
observations	TEXT	NULLABLE	---	Quality observations/notes
result	ENUM(...)	NOT NULL	---	Pass / Fail / Rework
check_date	DATE	NOT NULL	---	Date of in-process check

7.4 qc_pdi (Pre-Dispatch Inspection)

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	PDI record ID
FK so_no	VARCHAR(30)	NOT NULL	FK → sale_orders.so_no	Sale order for dispatch
FK dispatch_id	VARCHAR(30)	NULLABLE	FK → dispatch_advice.dispatch_id	Dispatch advice
box_no	VARCHAR(50)	NOT NULL	---	Box/carton number
packaging_condition	ENUM(...)	NOT NULL	---	Good / Damaged
label_accuracy	ENUM(...)	NOT NULL	---	Correct / Incorrect
inspection_date	DATE	NOT NULL	---	Date of pre-dispatch check

7.5 qc_qrd (Quality Rejection Decision)

Column Name	Data Type	Constraint	References	Description
PK rejection_id	VARCHAR(30)	PK, NOT NULL	---	Rejection decision record
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Rejected item
source_module	VARCHAR(50)	NOT NULL	---	IQC / PQC / PDI / External_IQC
qty	DECIMAL(10, 2)	NOT NULL	---	Rejected quantity
action	ENUM(...)	NOT NULL	---	Scrap / Return / Rework / Downgrade
decision_date	DATE	NOT NULL	---	Date decision was taken
decision_by	VARCHAR(150)	NOT NULL	---	Person who decided

Module 8. Contractor Employee Management

All contractor_* tables now reference contractor_master (M.4) instead of vendor_master, cleanly separating raw-material suppliers from job-work contractors.

8.1 contractor_employee_master

CHANGED: contractor_id FK now references contractor_master.contractor_id.

Column Name	Data Type	Constraint	References	Description
PK worker_id	VARCHAR(20)	PK, NOT NULL	---	Contractor worker ID
FK contractor_id	BIGINT UNSIGNED	NOT NULL	FK → contractor_master.contractor_id	Contractor firm
worker_name	VARCHAR(150)	NOT NULL	---	Worker full name
skill_level	ENUM(...)	NOT NULL	---	Skilled / Unskilled / Semi-Skilled
aadhar_no	VARCHAR(20)	NOT NULL, UNIQUE	---	Aadhar card number
is_active	TINYINT(1)	DEFAULT 1	---	Active/Inactive

8.2 contractor_salary_head_master

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Rate head identifier
role	VARCHAR(100)	NOT NULL, UNIQUE	---	e.g. Helper, Welder, Supervisor
daily_rate	DECIMAL(10, 2)	NOT NULL	---	Daily wage rate
overtime_rate	DECIMAL(10, 2)	NOT NULL	---	Overtime rate per hour

8.3 contractor_salary_structure

NEW: Maps each worker to their applicable salary head / rate. Mirrors the pattern of employee_salary_structure (5.3) for contractors.

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Structure record ID
FK worker_id	VARCHAR(20)	NOT NULL	FK → contractor_employee_master.worker_id	Worker
FK salary_head_id	BIGINT UNSIGNED	NOT NULL	FK → contractor_salary_head_master.id	Rate head applied
effective_date	DATE	NOT NULL	---	Date this rate became effective

8.4 contractor_salary_sheet

CHANGED: contractor_id FK now references contractor_master.contractor_id.

Column Name	Data Type	Constraint	References	Description
-------------	-----------	------------	------------	-------------

PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Payout record ID
FK worker_id	VARCHAR(20)	NOT NULL	FK → contractor_employ ee_master.worker_id	Worker
FK contractor_id	BIGINT UNSIGNED	NOT NULL	FK → contractor_master.contractor_id	Contractor firm
month	TINYINT	NOT NULL	---	Payroll month (1-12)
year	YEAR	NOT NULL	---	Payroll year
days_worked	DECIMAL(5,2)	NOT NULL	---	Days attended
overtime_hours	DECIMAL(6,2)	DEFAULT 0	---	Overtime hours worked
total_payable	DECIMAL(12,2)	GENERATED ALWAYS	---	(days × daily_rate) + (OT_hours × OT_rate)

8.5 contractor_advance_memo

CHANGED: contractor_id FK now references contractor_master.contractor_id.

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Advance record ID
FK contractor_id	BIGINT UNSIGNED	NOT NULL	FK → contractor_master.contractor_id	Contractor firm
advance_date	DATE	NOT NULL	---	Date advance was given
amount	DECIMAL(12,2)	NOT NULL	---	Advance amount
remarks	TEXT	NULLABLE	---	Reason for advance

8.6 contractor_voucher_payment

CHANGED: contractor_id FK now references contractor_master. Added finance_ledger_id FK → financial_transaction_ledger. On payment, a ledger entry is created (Debit: Contractor Payable, Credit: Bank; TDS Payable credited separately), connecting contractor payments to Finance books.

Column Name	Data Type	Constraint	References	Description
PK voucher_no	VARCHAR(30)	PK, NOT NULL	---	Payment voucher number
FK contractor_id	BIGINT UNSIGNED	NOT NULL	FK → contractor_master.contractor_id	Contractor paid
FK salary_sheet_ref	BIGINT UNSIGNED	NULLABLE	FK → contractor_salary_sheet.id	Salary sheet record
payment_date	DATE	NOT NULL	---	Date of payment
gross_amount	DECIMAL(14,2)	NOT NULL	---	Gross payable before TDS
tds_deducted	DECIMAL(12,2)	DEFAULT 0	---	TDS amount deducted (per tds_section in contractor_master)

net_amount_paid	DECIMAL(14, 2)	NOT NULL	---	Net amount paid after TDS
FK finance_ledger_id	BIGINT UNSIGNED	NOT NULL	FK → financial_transaction_ledger.ledger_id	Auto-created Finance ledger entry

Module 9. Maintenance Management

9.1 tool_master

Column Name	Data Type	Constraint	References	Description
PK asset_code	VARCHAR(30)	PK, NOT NULL	---	Tool/asset code
tool_name	VARCHAR(200)	NOT NULL	---	Tool description
FK warehouse_id	BIGINT UNSIGNED	NULLABLE	FK → warehouse_master.warehouse_id	Storage location
location	VARCHAR(100)	NOT NULL	---	Rack/area location
maintenance_interval_days	SMALLINT	NOT NULL	---	Days between maintenance
next_maintenance_date	DATE	NULLABLE	---	Next scheduled maintenance

9.2 tool_maintenance_chart

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Chart entry ID
FK asset_code	VARCHAR(30)	NOT NULL	FK → tool_master.asset_code	Tool reference
scheduled_date	DATE	NOT NULL	---	Scheduled maintenance date
task_list	TEXT	NOT NULL	---	Tasks: Greasing, Cleaning, Inspection
status	ENUM(...)	DEFAULT "Pending"	---	Pending / Done / Overdue

9.3 tool_calibration_report

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Calibration report ID
FK asset_code	VARCHAR(30)	NOT NULL	FK → tool_master.asset_code	Tool reference

calibration_date	DATE	NOT NULL	---	Date of calibration
standard_value	VARCHAR(100)	NOT NULL	---	Standard/reference value
actual_value	VARCHAR(100)	NOT NULL	---	Actual measured value
result	ENUM(...)	NOT NULL	---	Pass / Fail

9.4 tool_maintenance_memo

Column Name	Data Type	Constraint	References	Description
PK job_id	VARCHAR(30)	PK, NOT NULL	---	Maintenance job ID
FK asset_code	VARCHAR(30)	NOT NULL	FK → tool_master.asset_code	Tool reference
issue_description	TEXT	NOT NULL	---	Problem / issue reported
spares_used	TEXT	NULLABLE	---	Spare parts consumed
cost	DECIMAL(10,2)	DEFAULT 0	---	Repair cost
technician	VARCHAR(150)	NOT NULL	---	Technician name
repair_date	DATE	NOT NULL	---	Date repair was done

Module 10. Stores / Warehouse Management

10.1 warehouse_master

Column Name	Data Type	Constraint	References	Description
PK warehouse_id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Warehouse identifier
warehouse_name	VARCHAR(200)	NOT NULL	---	Warehouse name
address	TEXT	NULLABLE	---	Physical address
manager_name	VARCHAR(150)	NULLABLE	---	Warehouse manager

10.2 warehouse_opening

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Opening stock entry
FK warehouse_id	BIGINT UNSIGNED	NOT NULL	FK → warehouse_master.warehouse_id	Warehouse
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.pr	Item

			oduct_id	
opening_qty	DECIMAL(12, 4)	NOT NULL	---	Opening quantity
opening_value	DECIMAL(14, 2)	NOT NULL	---	Opening value (per unit × qty)
entry_date	DATE	NOT NULL	---	Date of opening entry

10.3 dispatch_srv (Service Voucher / Returnable Gate Pass)

Column Name	Data Type	Constraint	References	Description
PK srv_no	VARCHAR(30)	PK, NOT NULL	---	SRV number
srv_date	DATE	NOT NULL	---	Issue date
party_name	VARCHAR(200)	NOT NULL	---	Recipient party name
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.pr oduct_id	Item dispatched
qty	DECIMAL(10, 2)	NOT NULL	---	Quantity dispatched
is_returnable	TINYINT(1)	DEFAULT 1	---	1=Returnable gate pass
return_expected_date	DATE	NULLABLE	---	Expected return date

10.4 stock_transfers

Column Name	Data Type	Constraint	References	Description
PK transfer_id	VARCHAR(30)	PK, NOT NULL	---	Transfer identifier
FK from_warehouse_id	BIGINT UNSIGNED	NOT NULL	FK → warehouse_maste r.warehouse_id	Source warehouse
FK to_warehouse_id	BIGINT UNSIGNED	NOT NULL	FK → warehouse_maste r.warehouse_id	Destination warehouse
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.pr oduct_id	Item transferred
qty	DECIMAL(10, 2)	NOT NULL	---	Quantity transferred
transfer_date	DATE	NOT NULL	---	Date of transfer

10.5 warehouse_material_receipt

Column Name	Data Type	Constraint	References	Description
PK receipt_id	VARCHAR(30)	PK, NOT NULL	---	Receipt identifier
source_doc_ref	VARCHAR(50)	NOT NULL	---	Source document (SRV/Transfer ID)
FK product_id	BIGINT	NOT NULL	FK →	Item received

	UNSIGNED		product_master.pr oduct_id	
qty_received	DECIMAL(10, 2)	NOT NULL	---	Quantity received
received_date	DATE	NOT NULL	---	Date received

Module 11. Asset Management

11.1 fix_asset_master

Column Name	Data Type	Constraint	References	Description
PK asset_tag	VARCHAR(30)	PK, NOT NULL	---	Asset tag number
asset_name	VARCHAR(20 0)	NOT NULL	---	Asset description
asset_group	ENUM(...)	NOT NULL	---	IT / Plant / Furniture / Vehicle
purchase_date	DATE	NOT NULL	---	Date of purchase
purchase_value	DECIMAL(14, 2)	NOT NULL	---	Original cost of asset
is_active	TINYINT(1)	DEFAULT 1	---	In service flag

11.2 asset_addition_memo

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREM ENT	---	Addition memo ID
FK asset_tag	VARCHAR(30)	NOT NULL	FK → fix_asset_master. asset_tag	Asset
invoice_ref	VARCHAR(10 0)	NULLABLE	---	Purchase invoice reference
installation_date	DATE	NOT NULL	---	Date installed/commissioned
depreciation_rate	DECIMAL(5,2)	NOT NULL	---	Annual depreciation %

11.3 asset_allocation_master

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREM ENT	---	Allocation record ID
FK asset_tag	VARCHAR(30)	NOT NULL	FK → fix_asset_master. asset_tag	Asset allocated
FK emp_code	VARCHAR(20)	NOT NULL	FK → employee_master. emp_code	Employee receiving asset
FK dept_id	BIGINT	NOT NULL	FK →	Department

	UNSIGNED		department_master.dept_id	
date_assigned	DATE	NOT NULL	---	Date asset was assigned
date_returned	DATE	NULLABLE	---	Date asset was returned

11.4 asset_sale_memo

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Sale memo ID
FK asset_tag	VARCHAR(30)	NOT NULL	FK → fix_asset_master.asset_tag	Asset sold
sale_date	DATE	NOT NULL	---	Date of sale/disposal
sale_value	DECIMAL(14, 2)	NOT NULL	---	Actual sale proceeds
book_value	DECIMAL(14, 2)	NOT NULL	---	Net book value at time of sale

11.5 asset_depreciation_voucher

CHANGED: Added ledger_id FK → financial_transaction_ledger. Year-end depreciation is posted as a ledger entry (Debit: Depreciation Expense, Credit: Accumulated Depreciation), completing the Finance linkage.

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Depreciation entry ID
FK asset_tag	VARCHAR(30)	NOT NULL	FK → fix_asset_master.asset_tag	Asset
financial_year	VARCHAR(10)	NOT NULL	---	e.g. 2024-25
opening_balance	DECIMAL(14, 2)	NOT NULL	---	WDV at start of year
depreciation_amount	DECIMAL(14, 2)	NOT NULL	---	Depreciation for year
closing_balance	DECIMAL(14, 2)	GENERATED ALWAYS	---	opening_balance – depreciation_amount
FK ledger_id	BIGINT UNSIGNED	NULLABLE	FK → financial_transaction_ledger.ledger_id	Finance ledger entry

Module 12. Statutory Management (Taxation)

All statutory tables now draw source data from the relevant operational modules to avoid duplication. GST data is sourced from invoices and purchase_billbook; TDS from voucher_payments and contractor_voucher_payment; balance sheet from financial_transaction_ledger.

12.1 gst_taxation_master

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	GST rule ID
hsn_code	VARCHAR(20)	NOT NULL, UNIQUE	---	HSN/SAC code
description	VARCHAR(300)	NOT NULL	---	Goods/service description
igst_percent	DECIMAL(5,2)	DEFAULT 0	---	IGST rate
cgst_percent	DECIMAL(5,2)	DEFAULT 0	---	CGST rate
sgst_percent	DECIMAL(5,2)	DEFAULT 0	---	SGST rate
effective_from	DATE	NOT NULL	---	Rate effective date

12.2 gstr1_upload

CHANGED: invoice_no FK already links to invoices. System auto-populates GSTIN, taxable value, and tax from invoices at export time — no manual re-entry required.

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	GSTR-1 entry ID
month	VARCHAR(10)	NOT NULL	---	YYYY-MM filing period
FK invoice_no	VARCHAR(30)	NOT NULL	FK → invoices.invoice_no	Source sales invoice
customer_gstin	VARCHAR(20)	NULLABLE	---	Populated from customer_master
taxable_value	DECIMAL(14,2)	NOT NULL	---	Populated from invoices.taxable_value
tax_amount	DECIMAL(14,2)	NOT NULL	---	Populated from invoices.gst_amount
state	VARCHAR(10)	NOT NULL	---	Place of supply state
transaction_type	ENUM(...)	NOT NULL	---	B2B / B2C / Export

12.3 gst2a_reconciliation

CHANGED: Added vendor_id and bill_id FKs. This allows the system to auto-pull the company's purchase GST records from purchase_billbook and compare against the portal 2A data, flagging mismatches automatically.

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Reconciliation entry ID
month	VARCHAR(10)	NOT NULL	---	YYYY-MM filing period
FK vendor_id	BIGINT UNSIGNED	NOT NULL	FK → vendor_master.vendor_id	Vendor GSTIN source
FK bill_id	VARCHAR(30)	NULLABLE	FK →	Purchase bill being reconciled

	)		purchase_billbook .bill_id	
vendor_gstin	VARCHAR(20)	NOT NULL	---	Vendor GSTIN (from vendor_master)
total_itc_claimed	DECIMAL(14, 2)	NOT NULL	---	Input Tax Credit as per books
portal_itc_available	DECIMAL(14, 2)	NOT NULL	---	ITC available in GSTR-2A portal
matched_amount	DECIMAL(14, 2)	NOT NULL	---	Matched ITC
mismatch_amount	DECIMAL(14, 2)	GENERATED ALWAYS	---	total_itc_claimed – matched_amount

12.4 gst_deposit_challan

CHANGED: Added journal_voucher_gst_id and ledger_id FKs. GST payment challans now trace back to the specific GST journal entry (4.4) and also post to the financial_transaction_ledger (Debit: GST Payable, Credit: Bank).

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Challan entry ID
challan_no	VARCHAR(50)	NOT NULL, UNIQUE	---	GST challan number
cpin	VARCHAR(20)	NOT NULL	---	CPIN from GST portal
challan_date	DATE	NOT NULL	---	Payment date
bank	VARCHAR(150)	NOT NULL	---	Bank used for payment
tax_type	ENUM(...)	NOT NULL	---	CGST / SGST / IGST / CESS
amount	DECIMAL(14, 2)	NOT NULL	---	Amount deposited
FK journal_voucher_gst_id	BIGINT UNSIGNED	NULLABLE	FK → journal_voucher_gst.id	Source GST journal entry (Module 4.4)
FK ledger_id	BIGINT UNSIGNED	NOT NULL	FK → financial_transaction_ledger.ledger_id	Finance ledger entry (Debit: GST Payable, Credit: Bank)

12.5 tds_trace_details

CHANGED: Added voucher_payment_no and contractor_payment_no FKs. TDS deducted on vendor payments (2.10) and contractor payments (8.6) is now traceable directly from TDS records, enabling accurate TDS certificates.

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	TDS trace entry ID
tds_section	VARCHAR(20)	NOT NULL	---	e.g. 194C, 194J
deductee_name	VARCHAR(200)	NOT NULL	---	Party from whom TDS is deducted
payment_amount	DECIMAL(14, 2)	NOT NULL	---	Total payment on which TDS

	2)			applies
tds_rate	DECIMAL(5,2)	NOT NULL	---	TDS rate percentage
tds_amount	DECIMAL(14,2)	NOT NULL	---	TDS amount deducted
tds_certificate_no	VARCHAR(50)	NULLABLE	---	Form 16A certificate number
deduction_date	DATE	NOT NULL	---	Date of TDS deduction
FK voucher_payment_no	VARCHAR(30)	NULLABLE	FK → voucher_payment.s.voucher_no	Source purchase payment (if vendor)
FK contractor_payment_no	VARCHAR(30)	NULLABLE	FK → contractor_voucher_payment.voucher_no	Source contractor payment (if contractor)

12.6 tcs_details

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	TCS entry ID
FK invoice_no	VARCHAR(30)	NOT NULL	FK → invoices.invoice_no	Source sales invoice
customer_name	VARCHAR(200)	NOT NULL	---	Populated from customer_master
sale_value	DECIMAL(14,2)	NOT NULL	---	Gross sale value
tcs_rate	DECIMAL(5,2)	NOT NULL	---	TCS rate e.g. 0.1%
tcs_amount	DECIMAL(12,2)	NOT NULL	---	TCS amount collected

12.7 gstr1_gstr2_register

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Register entry ID
date_from	DATE	NOT NULL	---	Period start date
date_to	DATE	NOT NULL	---	Period end date
register_type	ENUM(...)	NOT NULL	---	GSTR1 / GSTR2
transaction_type	ENUM(...)	NOT NULL	---	B2B / B2C / Export / Import
total_tax_liability	DECIMAL(16,2)	NOT NULL	---	Total tax for period

12.8 cheque_book_management

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Cheque book ID

		ENT		
bank_account	VARCHAR(150)	NOT NULL	---	Bank account reference
start_leaf_no	VARCHAR(20)	NOT NULL	---	First cheque leaf number
end_leaf_no	VARCHAR(20)	NOT NULL	---	Last cheque leaf number
issued_date	DATE	NOT NULL	---	Date cheque book received

12.8a cheque_leaf_status

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Leaf status record
FK cheque_book_id	BIGINT UNSIGNED	NOT NULL	FK → cheque_book_management.id	Parent cheque book
leaf_no	VARCHAR(20)	NOT NULL	---	Individual cheque leaf number
status	ENUM(...)	NOT NULL	---	Blank / Used / Cancelled
issued_to	VARCHAR(200)	NULLABLE	---	Party name cheque was issued to
issue_date	DATE	NULLABLE	---	Date cheque was issued

12.9 balance_sheet

CHANGED: Added finance_period reference. Balance sheet is now derived from financial_transaction_ledger aggregations, not manually entered, ensuring accuracy.

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Balance sheet entry ID
as_on_date	DATE	NOT NULL	---	Balance sheet date
financial_year	VARCHAR(10)	NOT NULL	---	e.g. 2024-25
total_assets	DECIMAL(16, 2)	NOT NULL	---	Computed from ledger asset accounts
total_liabilities	DECIMAL(16, 2)	NOT NULL	---	Computed from ledger liability accounts
capital_account	DECIMAL(16, 2)	NOT NULL	---	Owner's equity / capital
current_assets	DECIMAL(16, 2)	NOT NULL	---	Cash + Receivables + Stock
current_liabilities	DECIMAL(16, 2)	NOT NULL	---	Payables + Outstanding expenses
net_profit_loss	DECIMAL(16, 2)	NOT NULL	---	Derived from P&L ledger aggregation
generated_by	BIGINT UNSIGNED	NULLABLE	---	FK → users.id — who generated

Module 13. Production Simulation & Forecasting

13.1 master_production_schedule

Column Name	Data Type	Constraint	References	Description
PK mps_id	VARCHAR(30)	PK, NOT NULL	---	MPS batch ID
created_by	BIGINT UNSIGNED	NOT NULL	FK → users.id	User who created MPS
created_date	DATE	NOT NULL	---	MPS creation date
shift_hours	DECIMAL(5,2)	DEFAULT 10	---	Working hours per shift
worker_count	SMALLINT	NOT NULL	---	Number of workers available
created_at	TIMESTAMP	DEFAULT NOW()	---	Creation timestamp

13.2 mps_items

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	MPS line item ID
FK mps_id	VARCHAR(30)	NOT NULL	FK → master_production_schedule.mps_id	MPS batch
FK product_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	Target product
target_quantity	DECIMAL(10,2)	NOT NULL	---	Planned production quantity

13.3 mrp_output

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	MRP result ID
FK mps_id	VARCHAR(30)	NOT NULL	FK → master_production_schedule.mps_id	Source MPS
FK raw_material_id	BIGINT UNSIGNED	NOT NULL	FK → product_master.product_id	RM required
required_qty	DECIMAL(12,4)	NOT NULL	---	Total RM needed
current_stock	DECIMAL(12,4)	NOT NULL	---	Stock at time of simulation (from stock_master)
shortage_flag	TINYINT(1)	GENERATED ALWAYS	---	1 if required > current stock

13.4 crp_output

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	CRP result ID
FK mps_id	VARCHAR(30)	NOT NULL	FK → master_production_schedule.mps_id	Source MPS
total_man_hours	DECIMAL(12,4)	NOT NULL	---	Total man-hours required (from production_bom)
days_required	DECIMAL(8,2)	NOT NULL	---	Days to complete production
estimated_completion_date	DATE	NOT NULL	---	Projected finish date
overload_alert	TINYINT(1)	DEFAULT 0	---	1 = Cannot finish within month

13.5 cost_estimation

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Cost estimate ID
FK mps_id	VARCHAR(30)	NOT NULL	FK → master_production_schedule.mps_id	Source MPS
labor_cost	DECIMAL(16,2)	NOT NULL	---	Total labour cost
overhead_electricity	DECIMAL(16,2)	NOT NULL	---	Electricity/overhead cost
material_cost	DECIMAL(16,2)	NOT NULL	---	Total raw material cost
total_cost	DECIMAL(16,2)	GENERATED ALWAYS	---	Sum of all cost components

13.6 resource_master

Column Name	Data Type	Constraint	References	Description
PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Resource record
resource_name	VARCHAR(200)	NOT NULL	---	Labour type or machine name
resource_type	ENUM(...)	NOT NULL	---	Labor / Machine
cost_per_hour	DECIMAL(10,2)	NOT NULL	---	Hourly cost rate
power_consumption_kwh	DECIMAL(8,4)	DEFAULT 0	---	kWh per hour (machines only)

13.7 shift_master

Column Name	Data Type	Constraint	References	Description
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PK id	BIGINT UNSIGNED	PK, NOT NULL, AUTO_INCREMENT	---	Shift definition
shift_name	VARCHAR(50)	NOT NULL	---	e.g. Morning, Evening
shift_hours	DECIMAL(5,2)	NOT NULL	---	Default hours in this shift
start_time	TIME	NOT NULL	---	Shift start time
end_time	TIME	NOT NULL	---	Shift end time